

1st lecture (14/03/2024) - Online

- ★ **Suggested book:** Optical Fiber Communication - John M. Senior (3rd edition).
- ★ **Chapter-01 complete:** up to slide 21.
- ★ **Read from slide.** (Question solution part may cover entire slide)

2nd lecture (21/04/2024) - Offline

- ★ **Total internal reflection:** If the dielectric on the other side of the interface has a refractive index n_2 which is less than n_1 , then the refraction is such that the ray path in this lower index medium is at an angle ϕ_2 to the normal, where ϕ_2 is greater than ϕ_1 . The angles of incidence ϕ_1 and refraction ϕ_2 are related to each other and to the refractive indices of the dielectrics by Snell's law of refraction, which states that:

$$n_1 \sin \phi_1 = n_2 \sin \phi_2$$

$$\Rightarrow \frac{\sin \phi_1}{\sin \phi_2} = \frac{n_2}{n_1}$$

This is the limiting case of refraction and the angle of incidence is now known as the critical angle ϕ_c . Value of the critical angle is given by:

$$\Rightarrow \sin \phi_c = n_2/n_1$$

$$\Rightarrow \phi_c = \sin^{-1} \frac{n_2}{n_1}$$

- ★ **Acceptance angle:** θ_a is the maximum angle to the axis at which light may enter the fiber in order to be propagated, and is often referred to as the acceptance angle* for the fiber.

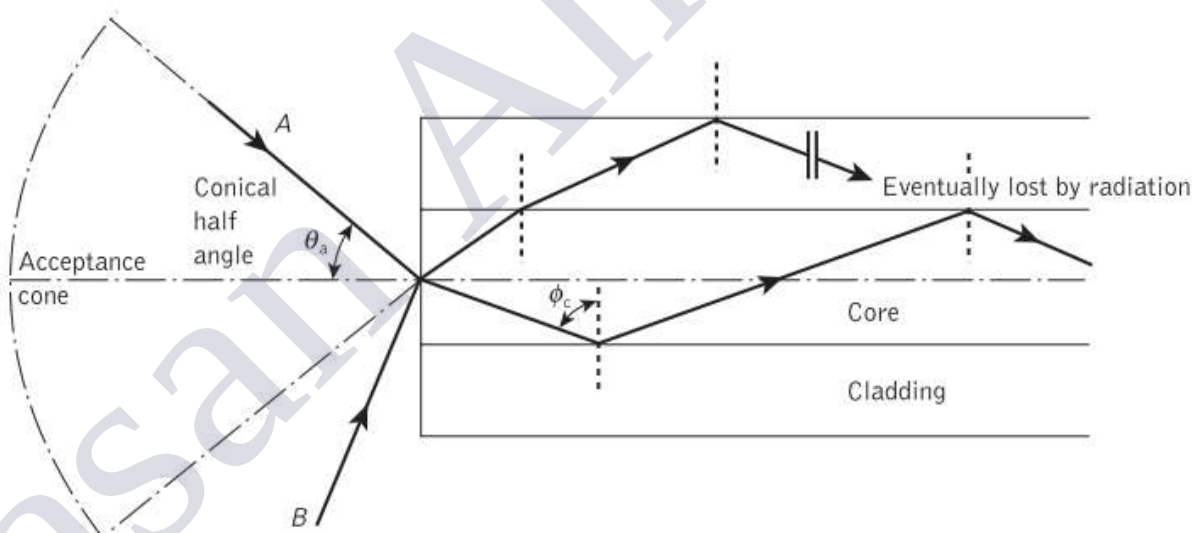


Figure 2.4 The acceptance angle θ_a when launching light into an optical fiber

☛ Numerical aperture:

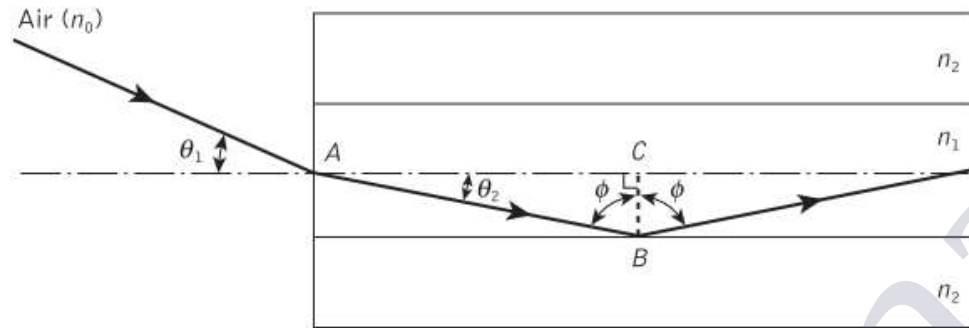


Figure 2.5 The ray path for a meridional ray launched into an optical fiber in air at an input angle less than the acceptance angle for the fiber

Assuming the entrance face at the fiber core to be normal to the axis, then considering the refraction at the air–core interface and using Snell’s law given by:

$$n_0 \sin \theta_1 = n_1 \sin \theta_2 \quad (2.3)$$

Considering the right-angled triangle ABC indicated in Figure 2.5, then:

$$\phi = \frac{\pi}{2} - \theta_2 \quad (2.4)$$

where ϕ is greater than the critical angle at the core–cladding interface. Hence Eq. (2.3) becomes:

$$n_0 \sin \theta_1 = n_1 \cos \phi \quad (2.5)$$

Using the trigonometrical relationship $\sin^2 \phi + \cos^2 \phi = 1$, Eq. (2.5) may be written in the form:

$$n_0 \sin \theta_1 = n_1 (1 - \sin^2 \phi)^{\frac{1}{2}} \quad (2.6)$$

When the limiting case for total internal reflection is considered, ϕ becomes equal to the critical angle for the core–cladding interface and is given by Eq. (2.2). Also in this limiting case θ_1 becomes the acceptance angle for the fiber θ_a . Combining these limiting cases into Eq. (2.6) gives:

$$n_0 \sin \theta_a = (n_1^2 - n_2^2)^{\frac{1}{2}} \quad (2.7)$$

Equation (2.7), apart from relating the acceptance angle to the refractive indices, serves as the basis for the definition of the important optical fiber parameter, the numerical aperture (NA). Hence the NA is defined as:

$$NA = n_0 \sin \theta_a = (n_1^2 - n_2^2)^{\frac{1}{2}} \quad (2.8)$$

The NA may also be given in terms of the relative refractive index difference Δ between the core and the cladding which is defined as:

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$$

$$\simeq \frac{n_1 - n_2}{n_1} \quad \text{for } \Delta \ll 1 \quad (2.9)$$

Hence combining Eq. (2.8) with Eq. (2.9) we can write:

$$NA = n_1(2\Delta)^{\frac{1}{2}} \quad (2.10)$$

➤ **Example-2.1:**

Example 2.1

A silica optical fiber with a core diameter large enough to be considered by ray theory analysis has a core refractive index of 1.50 and a cladding refractive index of 1.47.

Determine: (a) the critical angle at the core-cladding interface; (b) the NA for the fiber; (c) the acceptance angle in air for the fiber.

Solution: (a) The critical angle ϕ_c at the core-cladding interface is given by Eq. (2.2) where:

$$\phi_c = \sin^{-1} \frac{n_2}{n_1} = \sin^{-1} \frac{1.47}{1.50}$$

$$= 78.5^\circ$$

(b) From Eq. (2.8) the NA is:

$$NA = (n_1^2 - n_2^2)^{\frac{1}{2}} = (1.50^2 - 1.47^2)^{\frac{1}{2}}$$

$$= (2.25 - 2.16)^{\frac{1}{2}}$$

$$= 0.30$$

(c) Considering Eq. (2.8) the acceptance angle in air θ_a is given by:

$$\theta_a = \sin^{-1} NA = \sin^{-1} 0.30$$

$$= 17.4^\circ$$

➤ **Example-2.2:****Example 2.2**

A typical relative refractive index difference for an optical fiber designed for long-distance transmission is 1%. Estimate the NA and the solid acceptance angle in air for the fiber when the core index is 1.46. Further, calculate the critical angle at the core-cladding interface within the fiber. It may be assumed that the concepts of geometric optics hold for the fiber.

Solution: Using Eq. (2.10) with $\Delta = 0.01$ gives the NA as:

$$NA = n_1(2\Delta)^{\frac{1}{2}} = 1.46(0.02)^{\frac{1}{2}} \\ = 0.21$$

For small angles the solid acceptance angle in air ζ is given by:

$$\zeta \approx \pi\theta_a^2 = \pi \sin^2 \theta_a$$

Hence from Eq. (2.8):

$$\zeta \approx \pi(NA)^2 = \pi \times 0.04 \\ = 0.13 \text{ rad}$$

Using Eq. (2.9) for the relative refractive index difference Δ gives:

$$\Delta \approx \frac{n_1 - n_2}{n_1} = 1 - \frac{n_2}{n_1}$$

Hence

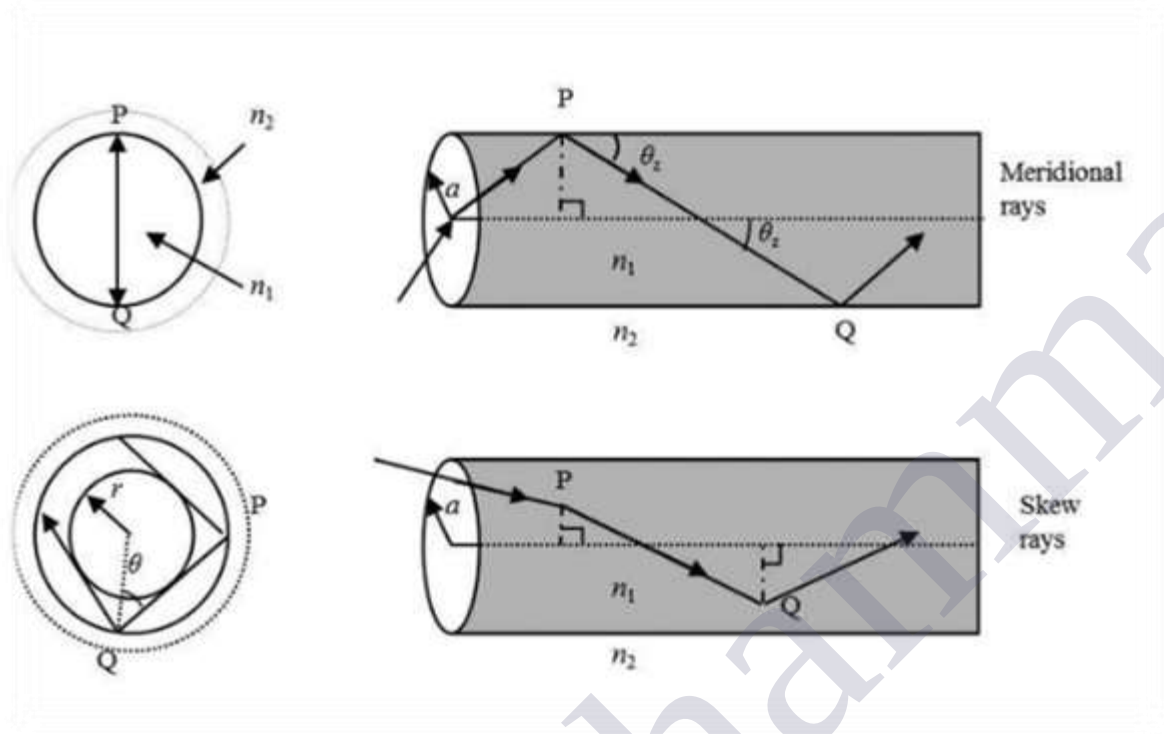
$$\frac{n_2}{n_1} = 1 - \Delta = 1 - 0.01 \\ = 0.99$$

From Eq. (2.2) the critical angle at the core-cladding interface is:

$$\phi_c = \sin^{-1} \frac{n_2}{n_1} = \sin^{-1} 0.99$$

81.9

☛ Meridional rays and Skew rays:



The acceptance conditions for skew rays are:

$$n_0 \sin \theta_{as} \cos \gamma = (n_1^2 - n_2^2)^{\frac{1}{2}} = NA \quad (2.16)$$

and in the case of the fiber in air ($n_0 = 1$):

$$\sin \theta_{as} \cos \gamma = NA \quad (2.17)$$

➤ Example-2.3:

Example 2.3

An optical fiber in air has an NA of 0.4. Compare the acceptance angle for meridional rays with that for skew rays which change direction by 100° at each reflection.

Solution: The acceptance angle for meridional rays is given by Eq. (2.8) with $n_0 = 1$ as:

$$\begin{aligned}\theta_s &= \sin^{-1} NA = \sin^{-1} 0.4 \\ &= 23.6^\circ\end{aligned}$$

The skew rays change direction by 100° at each reflection, therefore $\gamma = 50^\circ$. Hence using Eq. (2.17) the acceptance angle for skew rays is:

$$\begin{aligned}\theta_{as} &= \sin^{-1} \left(\frac{NA}{\cos \gamma} \right) = \sin^{-1} \left(\frac{0.4}{\cos 50^\circ} \right) \\ &= 38.5^\circ\end{aligned}$$

In this example, the acceptance angle for the skew rays is about 15° greater than the corresponding angle for meridional rays. However, it must be noted that we have only compared the acceptance angle of one particular skew ray path. When the light input to the fiber is at an angle to the fiber axis, it is possible that γ will vary from zero for meridional rays to 90° for rays which enter the fiber at the core-cladding interface giving acceptance of skew rays over a conical half angle of $\pi/2$ radians.

3rd lecture (12/05/2024) (Midterm syllabus ended here)-Online

- ★ Start Chapter-03 from slide.
- ★ Chapter ended (Theory part from slide).
- ★ Read from slide.

4th lecture (23/05/2024)-Offline

★ Example-3.1, 3.2, 3.4.

Example 3.1

When the mean optical power launched into an 8 km length of fiber is 120 μ W, the mean optical power at the fiber output is 3 μ W.

Determine:

- the overall signal attenuation or loss in decibels through the fiber assuming there are no connectors or splices;
- the signal attenuation per kilometer for the fiber.
- the overall signal attenuation for a 10 km optical link using the same fiber with splices at 1 km intervals, each giving an attenuation of 1 dB;
- the numerical input/output power ratio in (c).

Solution: (a) Using Eq. (3.1), the overall signal attenuation in decibels through the fiber is:

$$\begin{aligned}\text{Signal attenuation} &= 10 \log_{10} \frac{P_i}{P_o} = 10 \log_{10} \frac{120 \times 10^{-6}}{3 \times 10^{-6}} \\ &= 10 \log_{10} 40 = 16.0 \text{ dB}\end{aligned}$$

(b) The signal attenuation per kilometer for the fiber may be simply obtained by dividing the result in (a) by the fiber length which corresponds to it using Eq. (3.3) where:

$$\alpha_{\text{dB}} L = 16.0 \text{ dB}$$

hence:

$$\begin{aligned}\alpha_{\text{dB}} &= \frac{16.0}{8} \\ &= 2.0 \text{ dB km}^{-1}\end{aligned}$$

(c) As $\alpha_{\text{dB}} = 2 \text{ dB km}^{-1}$, the loss incurred along 10 km of the fiber is given by:

$$\alpha_{\text{dB}} L = 2 \times 10 = 20 \text{ dB}$$

However, the link also has nine splices (at 1 km intervals) each with an attenuation of 1 dB. Therefore, the loss due to the splices is 9 dB.

Hence, the overall signal attenuation for the link is:

$$\begin{aligned}\text{Signal attenuation} &= 20 + 9 \\ &= 29 \text{ dB}\end{aligned}$$

(d) To obtain a numerical value for the input/output power ratio, Eq. (3.2) may be used where:

$$\frac{P_i}{P_o} = 10^{29/10} = 794.3$$

Example 3.2

Silica has an estimated fictive temperature of 1400 K with an isothermal compressibility of $7 \times 10^{-11} \text{ m}^2 \text{ N}^{-1}$ [Ref. 13]. The refractive index and the photoelastic coefficient for silica are 1.46 and 0.286 respectively [Ref. 13]. Determine the theoretical attenuation in decibels per kilometer due to the fundamental Rayleigh scattering in silica at optical wavelengths of 0.63, 1.00 and 1.30 μm . Boltzmann's constant is $1.381 \times 10^{-23} \text{ J K}^{-1}$.

Solution: The Rayleigh scattering coefficient may be obtained from Eq. (3.4) for each wavelength. However, the only variable in each case is the wavelength, and therefore the constant of proportionality of Eq. (3.4) applies in all cases. Hence:

$$\begin{aligned}\gamma_R &= \frac{8\pi^3 n^8 p^2 \beta_e K T_f}{3\lambda^4} \\ &= \frac{248.15 \times 20.65 \times 0.082 \times 7 \times 10^{-11} \times 1.381 \times 10^{-23} \times 1400}{3 \times \lambda^4} \\ &= \frac{1.895 \times 10^{-28}}{\lambda^4} \text{ m}^{-1}\end{aligned}$$

At a wavelength of 0.63 μm :

$$\gamma_R = \frac{1.895 \times 10^{-28}}{0.158 \times 10^{-24}} = 1.199 \times 10^{-3} \text{ m}^{-1}$$

The transmission loss factor for 1 kilometer of fiber may be obtained using Eq. (3.5):

$$\begin{aligned}\mathcal{F}_{\text{km}} &= \exp(-\gamma_R L) = \exp(-1.199 \times 10^{-3} \times 10^3) \\ &= 0.301\end{aligned}$$

The attenuation due to Rayleigh scattering in decibels per kilometer may be obtained from Eq. (3.1) where:

$$\begin{aligned}\text{Attenuation} &= 10 \log_{10}(1/\mathcal{F}_{\text{km}}) = 10 \log_{10} 3.322 \\ &= 5.2 \text{ dB km}^{-1}\end{aligned}$$

At a wavelength of 1.0 μm :

$$\gamma_R = \frac{1.895 \times 10^{-28}}{10^{-24}} = 1.895 \times 10^{-4} \text{ m}^{-1}$$

Using Eq. (3.5):

$$\begin{aligned}\mathcal{F}_{\text{km}} &= \exp(-1.895 \times 10^{-4} \times 10^3) = \exp(-0.1895) \\ &= 0.827\end{aligned}$$

and Eq. (3.1):

$$\text{Attenuation} = 10 \log_{10} 1.209 = 0.8 \text{ dB km}^{-1}$$

At a wavelength of $1.30 \mu\text{m}$:

$$\gamma_R = \frac{1.895 \times 10^{-28}}{2.856 \times 10^{-24}} = 0.664 \times 10^{-4}$$

Using Eq. (3.5):

$$\mathcal{L}_{\text{km}} = \exp(-0.664 \times 10^{-4} \times 10^3) = 0.936$$

and Eq. (3.1):

$$\text{Attenuation} = 10 \log_{10} 1.069 = 0.3 \text{ dB km}^{-1}$$

The theoretical attenuation due to Rayleigh scattering in silica at wavelengths of 0.63, 1.00 and $1.30 \mu\text{m}$, from Example 3.2, is 5.2, 0.8 and 0.3 dB km^{-1} respectively. These theoretical results are in reasonable agreement with experimental work. For instance, a low reported value for Rayleigh scattering in silica at a wavelength of $0.6328 \mu\text{m}$ is 3.9 dB km^{-1} [Ref. 13]. However, values of 4.8 dB km^{-1} [Ref. 14] and 5.4 dB km^{-1} [Ref. 15] have also been reported. The predicted attenuation due to Rayleigh scattering against wavelength is indicated by a dashed line on the attenuation characteristics shown in Figures 3.1 and 3.3.

Example 3.4

Two step index fibers exhibit the following parameters:

- (a) a multimode fiber with a core refractive index of 1.500, a relative refractive index difference of 3% and an operating wavelength of 0.82 μm ;
- (b) an 8 μm core diameter single-mode fiber with a core refractive index the same as (a), a relative refractive index difference of 0.3% and an operating wavelength of 1.55 μm .

Estimate the critical radius of curvature at which large bending losses occur in both cases.

Solution: (a) The relative refractive index difference is given by Eq. (2.9) as:

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$$

Hence:

$$n_2^2 = n_1^2 - 2\Delta n_1^2 = 2.250 - 0.06 \times 2.250 \\ = 2.115$$

Using Eq. (3.8) for the multimode fiber critical radius of curvature:

$$R_c \simeq \frac{3n_1^2\lambda}{4\pi(n_1^2 - n_2^2)^{3/2}} = \frac{3 \times 2.250 \times 0.82 \times 10^{-6}}{4\pi \times (0.135)^{3/2}} \\ = 9 \mu\text{m}$$

(b) Again, from Eq. (2.9):

$$n_2^2 = n_1^2 - 2\Delta n_1^2 = 2.250 - (0.006 \times 2.250) \\ = 2.237$$

The cutoff wavelength for the single-mode fiber is given by Eq. (2.98) as:

$$\lambda_c = \frac{2\pi a n_1 (2\Delta)^{1/2}}{2.405} \\ = \frac{2\pi \times 4 \times 10^{-6} \times 1.500 (0.06)^{1/2}}{2.405} \\ = 1.214 \mu\text{m}$$

Substituting into Eq. (3.9) for the critical radius of curvature for the single-mode fiber gives:

$$R_{cs} \simeq \frac{20 \times 1.55 \times 10^{-6}}{(0.043)^{3/2}} \left(2.748 - \frac{0.996 \times 1.55 \times 10^{-6}}{1.214 \times 10^{-6}} \right)^{-3} \\ = 34 \text{ mm}$$